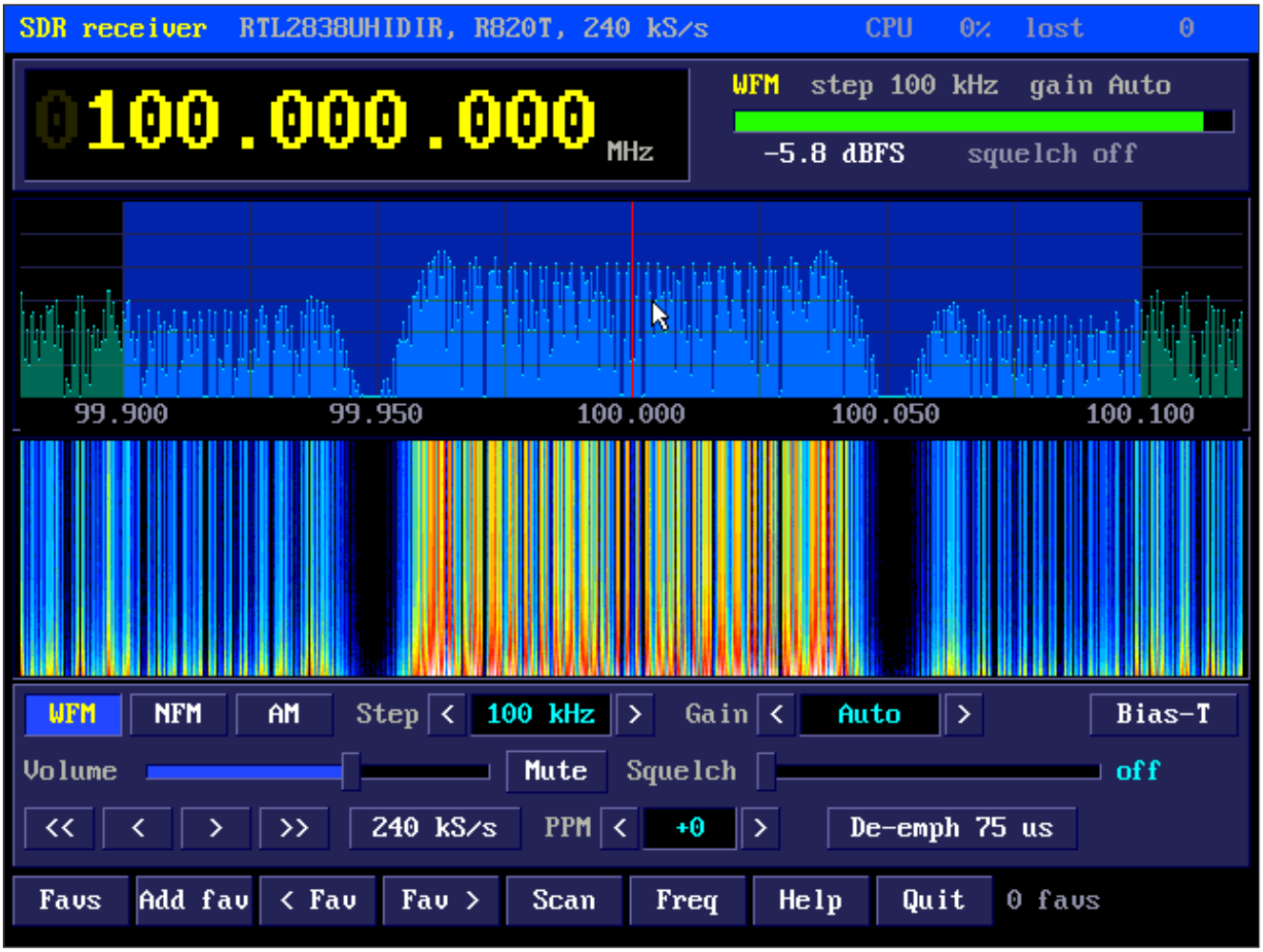
Pico 2 mini OS SDR Receiver

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A radio receiver for an RTL-SDR dongle plugged into the USB hub: it shows the spectrum and a waterfall of the band around the frequency tuned, and plays wide FM (broadcast), narrow FM (amateur, PMR, air band repeaters...) or AM through the HDMI sound, with a squelch, favourites and a scan of them.

What it needs

- An **RTL2832U dongle** with a Rafael Micro **R820T, R820T2 or R828D** tuner: the RTL-SDR Blog V3 and V4, the Nooelec NESDR range and most "RTL2832U + R820T2" DVB-T sticks. Dongles with the older Elonics E4000 or Fitipower FC0012/FC0013 tuners are recognised but not supported.
- An antenna on it. The telescopic dipole that comes with the Blog dongles does for broadcast FM and the 2 m band.
- The USB hub with its own 5 V supply (see Powering USB devices in the User's Manual): a dongle draws about 300 mA, more with the bias tee on.
- A USB mouse (the keys do everything too).

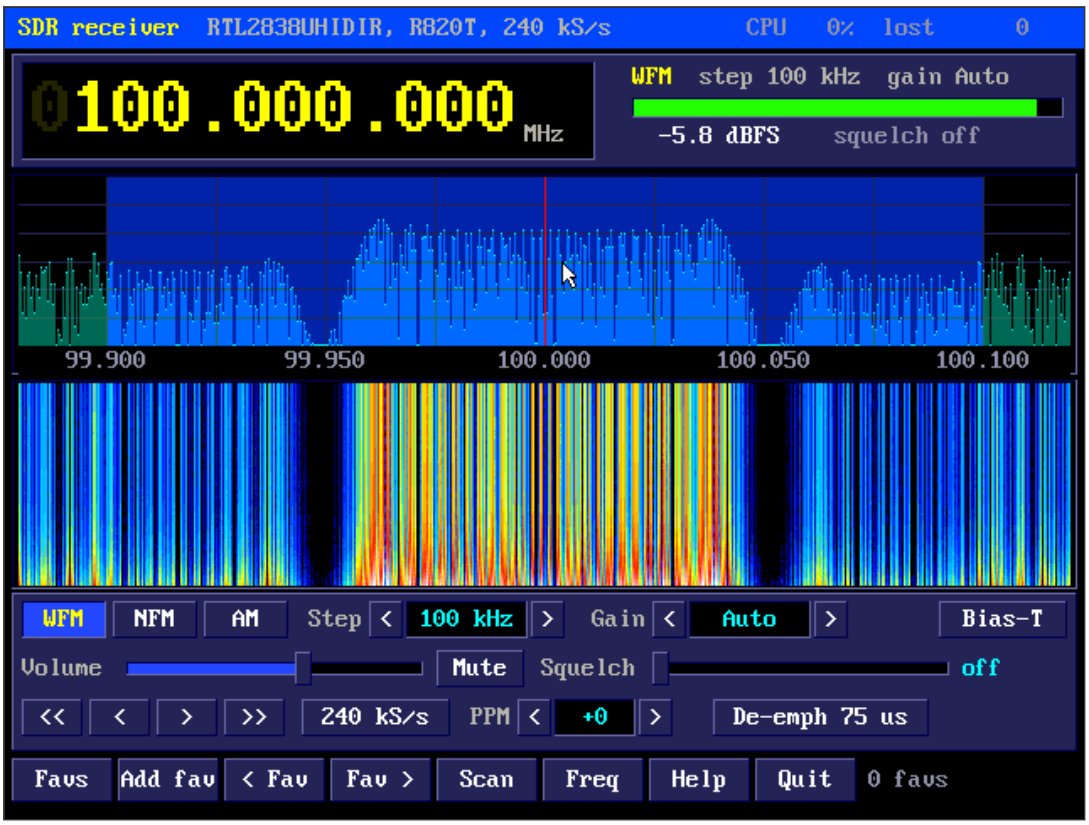
Plug the dongle in before or after starting `sdr`: the receiver starts it when it appears, and waits again if it is unplugged.

What it can do over the Pico's USB

The RP2350's USB runs at full speed (12 Mbit/s), and each sample from the dongle is two bytes, so the receiver runs the dongle at **240 kS/s** (480 KB/s; 288 kS/s can be chosen). That is a window **240 kHz wide** (288 kHz): enough for one broadcast FM station, in **mono** (its stereo and RDS signals reach past the edges of the window), or a stretch of narrow FM or AM channels. Computers run these dongles at 2 MS/s or more over high-speed USB; the Pico can't.

Frequencies from **0.5 MHz to 1766 MHz**. Below 24 MHz an R820T dongle (Blog V3) samples its antenna input directly, which works for strong signals (AM broadcast, shortwave) with a long wire; the **Blog V4** has an upconverter for HF that the receiver uses below 28.8 MHz.

The receiver

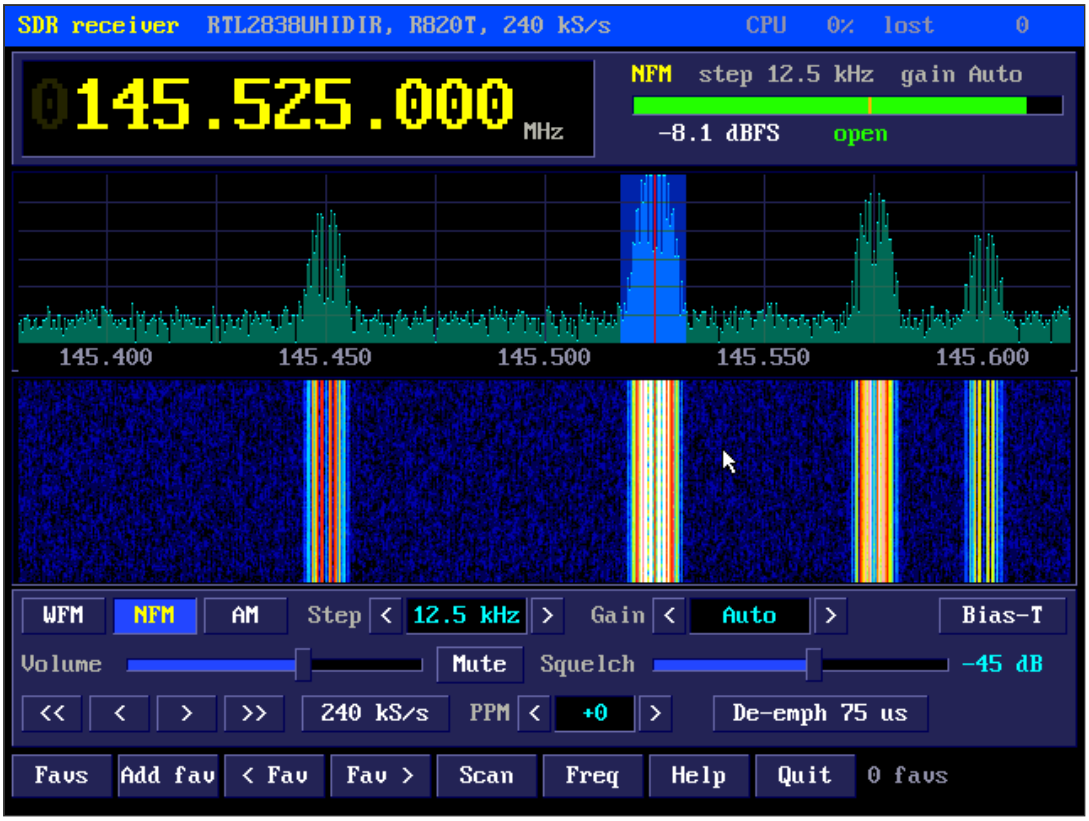


`sdr` opens it; `sdr 145.5 nfm` opens it on a frequency (MHz) and mode.

From the top:

- **The title strip:** the dongle's name and tuner, the sample rate (and "direct" when sampling the antenna directly); on the right, how much of the processor the demodulating takes and how many buffers of samples were lost (see Troubleshooting), or a message for a few seconds.
- **The frequency** in big digits, MHz.kHz.Hz. The wheel over a digit turns that digit up and down; a click on it adds one, a right click takes one away.
- Beside it: the **mode**, the **step** and the **gain**; the **signal meter** (the channel's level in dB below full scale, green while the squelch is open, with the squelch level marked in orange), and whether the squelch is open or shut. SCAN or MUTE shows there too.
- **The spectrum** of the window the dongle sends, with the frequency scale under it. The channel's band is shaded and the frequency tuned is the red line. Click anywhere to tune there (rounded to the step); the wheel over it tunes a step at a time. The scale follows the noise floor, so the bottom of the spectrum is always just under the noise.
- **The waterfall:** the spectrum going back about six seconds, newest at the top: dark blue is noise, then cyan, yellow, red and white as the signals get stronger. Clicks and the wheel work on it as on the spectrum.
- **The controls:** the modes (WFM, NFM, AM); the step and gain (the arrows, a click on the value, or the wheel over them); Bias-T; the volume and squelch sliders and Mute; `<<` `<` `>` `>>` (down a MHz, down a step, up a step, up a MHz); the sample rate; the crystal correction (PPM); and the wide FM de-emphasis.

- **The buttons along the bottom**, as in the sprite editor: Favs, Add fav, `< Fav` and `Fav >`, Scan, Freq (type a frequency), Help (the keys) and Quit; then how many favourites there are.



Modes

Mode	For	Channel
WFM	broadcast FM (88-108 MHz)	the whole window; mono, 15 kHz audio, de-emphasis 75 us (the Americas) or 50 us (Europe)
NFM	narrow FM: amateur, PMR446, business radio, marine...	15 kHz wide (12.5 and 25 kHz channels)
AM	AM broadcast, the air band (118-137 MHz), CB	10 kHz wide

In NFM and AM the channel can be anywhere in the window: tuning moves it there without retuning the dongle, until it would go more than about 75 kHz from the middle (the window's edges are less clean), when the dongle is tuned again so the channel is in the middle. In WFM the dongle is always tuned to the station (it fills the window).

Tuning

- Left and Right, `<` and `>`, or the wheel on the spectrum: a step down or up (to the nearest multiple of the step: from 145.512 a 12.5 kHz step up goes to 145.525).
- PgDn and PgUp, `<<` and `>>`: a megahertz.
- The wheel on a digit: that digit.
- A click on the spectrum or waterfall.

- Typing: the digits and a point start typing a frequency in MHz (it shows in place of the frequency, in cyan); Enter tunes there, Esc doesn't. Freq starts it with the mouse.

Steps: 100 Hz, 1, 5, 6.25, 9, 10, 12.5, 25, 50, 100 and 200 kHz, and 1 MHz. Broadcast FM is on 100 kHz (200 kHz in the Americas), the 2 m band on 12.5 or 25 kHz, AM broadcast on 9 kHz (10 kHz in the Americas), the air band on 25 kHz (8.33 kHz channels are 25 kHz apart in threes).

Gain

Auto lets the tuner set its own gain (its AGC), which suits most signals. Otherwise the tuner's 29 steps, 0 to 49.6 dB. More gain brings weak signals up, until strong ones overload the dongle: then false signals appear all over the spectrum, and less gain clears them. Below 24 MHz on an R820T dongle (direct sampling) there is no tuner gain.

Squelch

The squelch keeps the sound off until the channel's level rises above the squelch level, and closes it again 2 dB below. Drag the slider (or Q and Shift+Q) to just above the noise: the meter shows the level with nothing there, and the orange mark where the squelch is. All the way to the left it is off. The scan needs it.

Favourites

Add fav (F) keeps the frequency and mode, and asks for a name (Enter keeps it, Esc doesn't). **Favs** (L) lists them in frequency order: click one and **Tune** (or double click it, or Enter); **Rename**, **Delete** (or the Delete key), **Up** and **Down** to change their order, **Close** (Esc). `< Fav` and `Fav >` (P and N) tune to the previous and next ones. Up to 40.



Scan

Scan (C) goes round the favourites, listening to each for 0.15 s, and stops where the squelch opens; once the signal goes it waits 2 seconds and then goes on. Scan or C again stops it. It needs two favourites or more and the squelch on.

Other settings

- **Rate:** 240 or 288 kS/s. 288 shows a little more of the band, at 20% more USB traffic.
- **PPM:** the dongle's crystal is a few parts per million out (the Blog dongles less than 2; cheap ones up to 100). If stations show a little to one side of where they should be, tune to one whose frequency is known exactly and change the correction until it sits on the red line. At 100 MHz, 10 ppm is 1 kHz.
- **De-emph:** wide FM's de-emphasis, 75 us in the Americas and South Korea, 50 us elsewhere.
- **Bias-T:** puts 4.5 V up the antenna cable, for an LNA or active antenna that wants it (the Blog V3 and V4, and some Nooelec dongles, have one). Leave it off otherwise: some antennas short the cable for DC.

Everything is kept in `/SDR.INI` on the drive when the receiver closes (and when the favourites change).

Keys



Key	Does
Left / Right	down / up a step
PgDn / PgUp	down / up 1 MHz
Up / Down	volume
0-9 and .	type a frequency in MHz (Enter tunes, Esc doesn't, Backspace)
M	the next mode
S / Shift+S	a bigger / smaller step
G / Shift+G	more / less gain (below 0 dB: Auto)
Q / Shift+Q	squelch up / down
Space	mute
F / L	add a favourite / the list
N / P	the next / previous favourite
C	scan the favourites
F1, H or ?	the keys
Esc	close a box, stop typing, or leave

The receiver stops when it closes (the sound comes from its demodulating).

From the shell

```
sdr                       the receiver, where it was last
sdr 98.1                  on 98.1 MHz
sdr 145.5 nfm             on 145.500 MHz, narrow FM (wfm, nfm, am)
sdr info                  the dongle: its USB IDs and strings, its tuner
```

`sdr info` is a quick check that the dongle is seen and its tuner recognised:

```
USB 0bda:2838, 64-byte packets: RTLSDRBlog Blog V4
tuner R828D (RTL-SDR Blog V4); 0.5-1766 MHz, HF through its upconverter
```

Troubleshooting



- **"No RTL-SDR dongle plugged in"**: the dongle hasn't been found on the hub. When one is, the shell prints a line like `[USB: RTL-SDR dongle 0bda:2838, bulk endpoint 81, 64-byte packets]`. Dongles from other makers with IDs the driver doesn't know yet aren't taken: note its IDs so they can be added.
- **"Its E4000 tuner isn't supported"** (or FC0012, FC0013): an old dongle; only the R820T family is.
- **No sound but a signal on the spectrum:** the squelch is shut (the meter is grey), the volume is down, Mute is on, or the mode is wrong (wide FM sounds like silence or rushing in NFM, and NFM is quiet in WFM).
- **"lost" counts up** in the title strip: buffers of samples came while all eight were waiting, so the sound has gaps. It happens for a moment when the screen is drawn again; if it keeps counting, the demodulating is too slow (the CPU figure beside it) or USB is busy (copying files at the same time).
- **The CPU figure** is the share of core 0 the demodulating takes.
- **Stations are mirrored** (a station above the middle shows below it, and tuning to it finds nothing): the dongle's spectrum inversion setting is the wrong way round for it; report the dongle.

How it works

The RTL2832U was made for DVB-T television. In its "SDR mode" its ADC samples the tuner's output (a low IF of about 2 MHz), mixes it down to zero, filters and resamples it, and sends the 8-bit I and Q samples over USB. The R820T tuner before it mixes the antenna's frequency down to that IF, with a 350 kHz filter (its narrowest) that keeps out what the low sample rate would otherwise fold into the window.

USB. TinyUSB runs the USB host on core 1; a small class driver (`src/sdr/rtl_usb.c`) claims the dongle's first interface and its bulk endpoint (64-byte packets at full speed). The RP2350's host runs bulk transfers on the same hardware endpoint as control transfers (the patch in `src/usb_patch` that made the flash drive faster), so the dongle's setup and tuning requests, from core 0, go between its 2 KB sample transfers; while tuning, the stream is held so the requests run back to back. The samples come into a ring of eight 2 KB buffers in the apps' SRAM area (`app_overlay.h`), freed when the receiver closes.

The dongle (`src/sdr/rtl2832.c`): the RTL2832U's registers and the tuner's, over its I2C bus, set up as the open-source rtl-sdr driver does (osmocom's librtlsdr and the RTL-SDR Blog's version, which document how these chips work); the Blog V4 is recognised from its USB strings.

Demodulating (`src/sdr/sdr_dsp.c`, on core 0, in floating point):

- **WFM:** the phase change from each sample to the next (the FM discriminator) at 240 kS/s, then a 128-tap low-pass filter at 15.5 kHz that also takes out the 19 kHz stereo pilot, kept at every fifth sample for 48 kHz, and the de-emphasis.
- **NFM and AM:** the channel is shifted to zero (a rotating phasor), a 64-tap filter keeps it and every fifth (sixth at 288 kS/s) sample is kept for 48 kHz, then FM's discriminator and a 3 kHz low-pass, or AM's envelope divided by its own average (that is the AGC, and takes away the carrier).
- **The squelch** compares the channel's average power with its level.
- **The spectrum:** a 1024-point FFT with a Hann window, 20 times a second, smoothed; each of the 624 columns shows the strongest of the points under it. The waterfall is drawn in the framebuffer (moved down a row each time) and kept in PSRAM for redrawing.

The sound goes into a ring in PSRAM that core 1's HDMI sound plays at 48 kHz. The dongle's crystal and the Pico's don't agree exactly, so when the ring fills past three quarters a sample is dropped now and then; if it runs dry, playing waits until 50 ms have come again.

The PC test `tools/sdrtest/uitest.sh` runs the demodulators on made-up signals (FM and AM stations carrying a 1 kHz tone, beside other stations, with noise) and checks the tone comes out clean, then runs the receiver with a pretend dongle and makes these screenshots.